

Q18 /20

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/ 20

Completed

Microservice architecture provides many benefits over monolithic architecture. Which is/are the advantage(s) of microservice architecture? Select all correct options.

- ☐ A. Deployment in a microservice architecture is simpler because a microservice can be deployed for each service.
- ☐ B. Microservice architecture can reduce learning costs because it has a smaller codebase than monolithic systems.
- ☒ C. Microservice architecture can reduce single points of failure (SPOFs) because each service is loosely coupled.
- ☐ D. Microservice architecture can improve the total computational efficiency of the product because each service is minimal.

Answer >

JD

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered
- Q15 Not Answered
- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

Q17 /20

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/ 20

Completed

Choose the **two incorrect** options regarding SLOs (service level objectives) and SLAs (service level agreements) in the context of site reliability engineering.

- ☒ A. ☐ If an SLO is breached, there is a possibility of pursuing legal liability such as damages.
- ☒ B. ☐ Within the boundaries of meeting the SLO, the SRE team is allowed to intentionally cause outages if there are reasonable justifications.
- ☐ C. ☐ SLAs are usually set at values that are more difficult to achieve than SLOs.
- ☐ D. ☐ SLAs are established by considering factors such as the stability of competing services from other companies, the feasibility of service stability from an engineering perspective, and customer stability requirements.

Answer >

JD

Q6 Not Answered
Q7 Not Answered
Q8 Not Answered
Q9 Not Answered
Q10 Not Answered
Q11 Not Answered
Q12 Not Answered
Q13 Not Answered
Q14 Not Answered
Q15 Not Answered
Q16 Not Answered
Q17 Not Answered
Q18 Not Answered
Q19 Not Answered
Q20 Not Answered

< Q16 /20 >

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/ 20

Completed

Choose the description that is **incorrect** regarding serverless architecture.

- A. ☐ Applications deployed in serverless architecture start only when needed, automatically scale down during idle periods, and can reduce costs.
- B. ☐ Billing in serverless architecture typically adjusts automatically based on factors like the number of requests, and in some cases, there may be no charges when there are no requests.
- C. ☒ Even when instances scale up, serverless architecture has the feature of zero-second initialization, allowing the application to start processing requests instantly.
- D. ☐ While serverless architecture allows you to be server-agnostic, fine tuning low-level settings below the OS layer on the servers where applications run is often more challenging than using IaaS.

Answer >

Q6 Not Answered
Q7 Not Answered
Q8 Not Answered
Q9 Not Answered
Q10 Not Answered
Q11 Not Answered
Q12 Not Answered
Q13 Not Answered
Q14 Not Answered
Q15 Not Answered
📍 Q16 Not Answered
Q17 Not Answered
Q18 Not Answered
Q19 Not Answered
Q20 Not Answered

< **Q14** /20 >

raise is returned.

```
static boolean containConsecutive(int[] arr) {
    for (int i = 0; i < [ ① ]; i++) {
        if ([ ② ]) {
            return true;
        }
    }
    return false;
}
```

Choose the appropriate formula to enter into [①], [②] to make the function work.

- A. ☐ [①] arr.length-1, [②] arr[i] == arr[i-1]
- B. ☐ [①] arr.length-1, [②] arr[i] == arr[i+1]
- C. ☐ [①] arr.length, [②] arr[i] == arr[i-1]
- D. ☐ [①] arr.length, [②] arr[i] == arr[i+1]
- E. ☐ All the above are incorrect

Answer >

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/ 20

Completed

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- 📍 **Q14** Not Answered
- Q15 Not Answered
- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

Q13 /20

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/ 20

Completed

Beginner C

```
function sum3(a) {
  let answer = 0;
  let i = 0;
  while (1) {
    if (i == a.length) {
      break;
    }
    answer += a[i];
    i++;
  }
  return answer;
}
```

- A. ☐ Beginner A's function ✓
- B. ☐ Beginner B's function
- C. ☐ Beginner C's function ✓

Answer >

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered
- Q15 Not Answered
- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

Q2 /20

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/ 20

Completed

In UNIX-like OS, the authority for a certain file varies depending on the user who uses it.
Which of the following description is correct?

- A. ☐ There is one owner in the file, and authority is given only to that user.
- B. ☐ There are multiple owners in the file, and privileges are granted only to those users.
- C. ☒ There is one owner and group in the file, and different authorities are given to the user who is the owner, the users belonging to the group and users of the other type.
- D. ☐ There are multiple owner and group in the file, and two types of users, owner and users belonging to the group.

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q1 /20



Choose the DNS server setting that will prevent an existing host and its IP address from being known.

- A. ☐ Prohibit third-party mail relay.
- B. ☒ Control the accessing source of a zone transfer.
- C. ☐ Prohibit running a CGI program in other than a specified directory.
- D. ☐ Prohibit displaying file names existing in a directory.

[Answer >](#)

JD

Q1
Q2
Q3
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Q16
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Q18
Q19
Q20

Q1 /20



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/ 20

Completed

Choose the DNS server setting that will prevent an existing host and its IP address from being known.

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- B. ☒ Control the accessing source of a zone transfer.
- C. ☐ Prohibit running a CGI program in other than a specified directory.
- D. ☐ Prohibit displaying file names existing in a directory.

Answer >

Questions

- 📍 Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q3 /20 >

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/ 20

Completed

In UNIX-like OS, when trying to execute a command called foo in the /home/bar directory, the error -bash: foo: command not found was displayed. Which command is used to solve this error?

- ☒ A. export PATH=/home/bar:\$PATH
- B. ☐ export PWD=/home/bar
- C. ☐ export MANPATH=/home/bar:\$MANPATH
- D. ☐ export SHLV=2

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- 📍 Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q3 /20 >

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/ 20
Completed

In UNIX-like OS, when trying to execute a command called foo in the /home/bar directory, the error `-bash: foo: command not found` was displayed. Which command is used to solve this error?

- A. ☒ export PATH=/home/bar:\$PATH
- B. ☐ export PWD=/home/bar
- C. ☐ export MANPATH=/home/bar:\$MANPATH
- D. ☐ export SHLVL=2

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< **Q4** /20 >

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/ 20
Completed

Choose the public-key cryptosystem that utilizes the fact that the complexity does not fit within polynomial time.

- A. ☒ RSA
- B. ☐ IDEA
- C. ☐ DSA
- D. ☐ AES

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q3 / 30

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/ 30
Completed

Choose the correct description of database recovery processing.

- a. ☐ Restore using a journal file taken up during against a physical failure.
- b. ☐ Restore using a journal file after updating against a physical failure.
- c. ☐ Restore using a journal file taken up during against a logical failure.
- d. ☐ Restore using a journal file after updating against a logical failure.

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

◀ Q7 /20 ▶

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/ 20

Completed

What is it called when a specific character string is used as an input in the web application to illegally manipulate the server-side database.

- A. ☐ DoS attack
- B. ☐ DB attack
- C. ☐ SQL attack
- D. ☒ SQL injection

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q5 /20 >

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/ 20
Completed

When browsing a website (<https://example.com>), the contents displayed on the smartphone and the computer are different. Which statement below is **not** the cause?

- A. ☐ Because the content of the website is dynamically generated with JavaScript.
- B. ☐ Because the content of the website is dynamically switched based on the user agent.
- C. ☒ Because the content of the website was tampered by a communication path.
- D. ☐ Because the content of the website was implemented with responsive design.

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q6 /20

0

/ 20

Completed

Which TCP port number of the server the Web browser will connect to when specifying URL (Universal Resource Locator) as `http://ftp.example.jp/index.cgi?port=123` in the Web browser?

- A. ☐ 21
- B. ☒ 80
- C. ☐ 123
- D. ☐ 443

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q13 /20 >

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/ 20

Completed

Beginner B

```
function sum(a) {
  let answer = 0;
  for (let i = a.length; i > 0; i--) {
    answer += a[i];
  }
  return answer;
}
```

✗

Beginner C

```
function sum3(a) {
  let answer = 0;
  let i = 0;
  while (1) {
    if (i == a.length) {
      break;
    }
    answer += a[i];
    i++;
  }
  return answer;
}
```

✓

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- 📍 Q13 Not Answered
- Q14 Not Answered
- Q15 Not Answered
- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

Answer >

< Q11 /20 >

0
/ 20
Completed

There is a staircase of n steps and you are on the bottom of the staircase, on the 0-th step.

You can go up any of 1, 2, or 3 steps at a time.

Let us consider the problem of counting the number of step-taking patterns that will put you on exactly the n -th step from the bottom.

For instance, if there are $n = 4$ steps, there are 7 patterns as follows. Each number is the number of steps you take at one time.

- 1, 1, 1, 1.
- 2, 1, 1.
- 1, 2, 1.
- 1, 1, 2.
- 2, 2.
- 1, 3.
- 3, 1.

For this problem, you implemented a function `solve` as follows.
Note that all the variables are multiple precision integers.

```
function rec(i):
    if i == 0:
```

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q8 /20

0

/ 20
Completed

Choose the correct description of database rollforward processing.

- A. ☐ Restore using a journal file before updating against a physical failure.
- B. ☐ Restore using a journal file after updating against a physical failure.
- C. ☒ Restore using a journal file before updating against a logical failure.
- D. ☐ Restore using a journal file after updating against a logical failure.

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q9 /20

0
/ 20

Completed

For the two variables x and y, execute the following procedure from 1 to 4.
What is the value of x when processing is over?

1. Assign 2 to x and assign 3 to y.
2. Assign y by y minus 1.
3. Assign the sum of the value of x and the value of y into x.
4. If $y \neq 0$, return to step 2. If $y = 0$, the process is terminated.

Fill in the blanks of the following sentences with half-width numbers:

When processing is finished, the value of x becomes

5

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

Q10 /20

0
/ 20
Completed

Mr. Aoki is planning to visit the shrine for New Years. However, the parking lot at the shrine is very small. Once you park your car, you cannot leave until all of the cars that came after you leave. Choose the data structure that best represents the situation in this parking lot.

- A. ☒ stack
- B. ☐ queue
- C. ☐ heap
- D. ☐ linked list

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q12 /20 >

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/ 20
Completed

Your friend wrote the following function in Java. She says the function should return the average value of the given integer array `a`. Choose **all** of the correct statements about the function from the following.

```
static double mean(int[] a) {
    int answer = 0;
    for (int i = 0; i < a.length; i++) {
        answer += a[i];
    }
    return answer / a.length;
}
```

- A. ☐ If the length of `a` is 0, the function does not return any value.
- B. ☐ Aside from cases of integer overflow, the function always returns some value. However, the decimals of the returned value are rounded.
- C. ☐ An integer overflow may occur when `a` is long and has some large elements.
- D. ☐ None of the above are correct.

Answer >

Questions

- Q1 Not Answered
- Q2 Not Answered
- Q3 Not Answered
- Q4 Not Answered
- Q5 Not Answered
- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- 📍 Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered

< Q13 /20 >

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/ 20
Completed

You are teaching beginners A, B, C at your company. The beginners wrote the following functions. These functions receive an array of integers `a` and outputs the sum of all elements in array `a`.

Choose all functions that work properly. At least 1 function works properly.

Beginner A

```
function sum(a) {
  let answer = 0;
  for (let i = 0; i < a.length; i++) {
    answer += a[i];
  }
  return answer;
}
```

Beginner B

```
function sum(a) {
  let answer = 0;
  for (let i = a.length; i > 0; i--) {
    answer += a[i];
  }
  return answer;
}
```

Answer >

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
- Q14 Not Answered
- Q15 Not Answered
- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

< Q13 /20 >

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/ 20

Completed

You are teaching beginners A, B, C at your company. The beginners wrote the following functions. These functions receive an array of integers `a` and outputs the sum of all elements in array `a`.

Choose all functions that work properly. At least 1 function works properly.

Beginner A

```
function sum(a) {  
  let answer = 0;  
  for (let i = 0; i < a.length; i++) {  
    answer += a[i];  
  }  
  return answer;  
}
```

Beginner B

```
function sum(a) {  
  let answer = 0;  
  for (let i = a.length; i > 0; i--) {  
    answer += a[i];  
  }  
}
```

Answer >

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
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- Q14 Not Answered
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- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

< Q14 /20 >

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Completed

This Java function receives an input of array of integers `arr` and outputs `true` if there are any neighboring array elements that have the same value in the array.

For example:

Given `arr = {0, 3, 2, 2, 6, -1, 3}`, the elements `2, 2` are next to each other and are the same value, so `true` is returned.

Given `arr = {0, 3, 2, 5, 6, -1, 3}`, there are no elements with the same values next to each other, so `false` is returned.

```
static boolean containConsecutive(int[] arr) {  
    for (int i = 0; i < [ ① ]; i++) {  
        if ([ ② ]) {  
            return true;  
        }  
    }  
    return false;  
}
```

Choose the appropriate formula to enter into [①], [②] to make the function work.

A. ☐ [①] `arr.length-1`, [②] `arr[i] == arr[i-1]`

Answer >

- Q6 Not Answered
- Q7 Not Answered
- Q8 Not Answered
- Q9 Not Answered
- Q10 Not Answered
- Q11 Not Answered
- Q12 Not Answered
- Q13 Not Answered
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- Q16 Not Answered
- Q17 Not Answered
- Q18 Not Answered
- Q19 Not Answered
- Q20 Not Answered

< Q15 /20 >

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/ 20

Completed

Choose the description that is **incorrect** regarding the differences between in-memory databases and traditional databases like MySQL or PostgreSQL.

- A. ☐ In-memory databases are often faster compared to relational databases like MySQL.
- B. ☒ In-memory databases are resilient to the physical failure of an HDD or SSD, making data loss less likely compared to storage-based databases.
- C. ☐ In in-memory databases, ensuring sufficient main memory for storing all data is typically the only method used, rather than following rules like deleting infrequently used data to allocate enough capacity.
- D. ☐ Redis and Memcached are prominent examples of in-memory databases.

Answer >

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Q8 Not Answered
Q9 Not Answered
Q10 Not Answered
Q11 Not Answered
Q12 Not Answered
Q13 Not Answered
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Q17 Not Answered
Q18 Not Answered
Q19 Not Answered
Q20 Not Answered

Q19 /20

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/ 20

Completed

What are Docker images and Docker containers? Choose the correct option.

- A. ☐ Docker images are units of an OS, datafiles, and applications, including all dependencies between libraries used in the applications. A Docker container is an archive file that contains everything that can be used for Docker images.
- B. ☐ Docker containers are units of an OS, datafiles, and applications, including all dependencies between libraries used in the applications. A Docker image is an archive file that contains everything that can be used for Docker containers.
- C. ☐ Docker images are units of software that package code and all its dependencies. Docker containers are files used to create Docker images.
- ☒ Docker containers are units of software that package code and all its dependencies. Docker images are files used to create Docker containers.

Answer >

Q6 Not Answered
Q7 Not Answered
Q8 Not Answered
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Q20 Not Answered

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Completed

Which of the following is/are true about ChatOps? Choose all of the correct answers.

- ☒ It aims to create a centralized location where everyone's actions, tools, notifications, and diagnoses happen in full view.
- ☐ It primarily aims to expedite frequent communication, such as customer support, between information-technology operations (Ops) and end users using efficient chat tools.
- ☒ It contributes to creating a transparent team culture by bringing information that has been strongly dependent on a small set of employees to a centralized location.
- ☒ It is not only for technical teams but also for non-technical teams such as sales, marketing and finance. It can give a clear understanding of what is actually going on in the infrastructure and who is responsible for which systems and what each person does.

Answer >

Q6 Not Answered

Q7 Not Answered

Q8 Not Answered

Q9 Not Answered

Q10 Not Answered

Q11 Not Answered

Q12 Not Answered

Q13 Not Answered

Q14 Not Answered

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